To: Sarah Okafor, Head of E-Commerce | **From:** Amiya Sekhar, Data & ML Team | **Date:** 13 October 2026
Subject: Product recommendations and customer groups for Q4 planning

1. Summary

We built a recommendation model from two years of purchase history and tested it on the last two months of sales it had never seen. The honest headline: a plain best-sellers list still beats the model at getting at least one suggestion right (26.8% vs 21.4% of customers), but the model catches more of each customer's actual new purchases (2.4% vs 1.7%), does much better for light buyers, and gives every customer a personal list. We recommend using the model's "customers who bought this also bought" suggestions on product pages now, and splitting personalised lists: best sellers for heavy buyers, model suggestions for light and newer buyers.

2. What we built and why this method

We took the table of which customer bought which product (5,471 customers by 3,687 products) and compressed it into 50 "taste patterns" using a method called Truncated SVD. We chose SVD because purchase data only records what people bought: SVD treats "not bought" as "no information", while the usual alternative, PCA, treats it as a mild dislike. In testing the two performed almost the same; SVD was slightly better on every measure and simpler to run.

3. Size and speed

- Full purchase table: 161 MB. Compressed patterns: 3.7 MB.
- Whole nightly job (build the patterns, product-page suggestions, lists for all customers): under 1 second on a laptop.

4. What we lose by compressing

The 50 patterns keep about 38% of the detail in the purchase table. The rest is mostly individual one-off buying that no shared pattern explains, and it matters little for ranking suggestions, which is what we need.

5. Results on purchases the model never saw

Test: the last two months of data (9 October to 9 December 2011), 1,931 returning customers who bought something new, 10 suggestions each.

Measure	Best sellers	Our model
Hit rate: share of customers for whom at least one of the 10 suggestions was something they actually bought next	26.8%	21.4%
Share of each customer's new purchases caught by the 10 suggestions	1.7%	2.4%

- Best sellers win on "at least one right" because Q4 buying is dominated by popular Christmas stock that most customers pick up anyway.
- The model is better for light buyers. For customers with 1 to 10 products in their history, it caught 8.3% of their new purchases against 2.4% for best sellers.
- The model's lists are personal: 97.6% of customers share no product between their model list and the best-seller list.
- Product-page suggestions are coherent: Regency cakestand leads to Regency teacups, tea plates and teapot; red retrospot lunch bag leads to the other lunch bag designs.

6. Recommendation

- Use the model's "customers who bought this also bought" table on product pages now.
- Personalised lists: show best sellers to heavy buyers and model suggestions to light and newer buyers.
- Next step: our check found that counting purchases as simply "bought / not bought" raised the hit rate by about 2.4 points. Test that, plus a blend with popularity, on a fresh time period before switching.

7. Customer groups for marketing

Five groups from the same patterns (map: `reports/figures/ml6_map_k5_simple.png`). They describe buying tendencies, not hard boundaries.

- Occasional small buyers, 29.7%: few products, low spend, no distinctive items.
- Home décor & hearts, 23.9%: hanging hearts, zinc and ivory decorations, T-light holders.
- Kids' crafts & toys, 20.0%: paint sets, crayons, modelling clay, card games.
- Bulk bag buyers, 13.3%: jumbo and storage bags; highest spend, repeat trade-style orders.
- Baking & party, 13.2%: cake cases and paper doilies.

8. What could go wrong

- The test period was the Christmas peak; results may differ in other seasons.
- Rarely bought products can get odd matches because few customers bought them.
- 380 new customers had no history, so they can only get best sellers.
- We chose some settings after seeing test results, so the reported numbers are slightly optimistic.
- Very large one-off orders can skew a customer's suggestions; we dampen this, but it needs watching.